

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T
Semester IV) MERN Stack

Module - I

Practical – I

Roll No.: T007	Name: subhashree jena
Class: TYIT	Batch: 2
Date of Assignment: 20-06-206	Date/Time of Submission: 20-06-2026

1.Environment Setup and JavaScript Basics



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\IT-42\Desktop\MERN stack> node -v
v24.17.0
PS C:\Users\IT-42\Desktop\MERN stack> npm -v
11.13.0
```

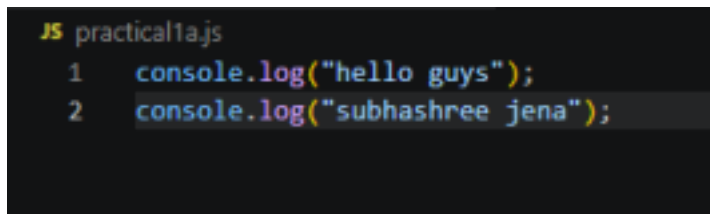
a. Install Node.js and Visual Studio Code, and verify installation.



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\IT-42\Desktop\MERN stack> node -v
v24.17.0
PS C:\Users\IT-42\Desktop\MERN stack> npm -v
11.13.0
```

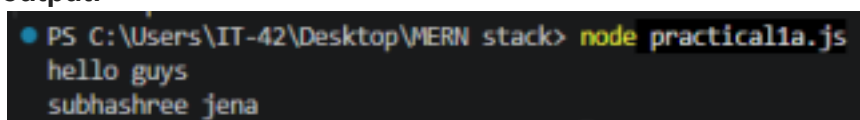
b. Create a simple Node.js file and run it using the terminal.

Code:-



```
JS practical1a.js
1 console.log("hello guys");
2 console.log("subhashree jena");
```

output:-



```
PS C:\Users\IT-42\Desktop\MERN stack> node practical1a.js
hello guys
subhashree jena
```

c.

Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

Code:-

```
JS practical1.js > ...
1
2   console.log("Welcome to Node.js!");
3
4
5   let num1 = 20;
6   let num2 = 5;
7
8   console.log("Number 1 =", num1);
9   console.log("Number 2 =", num2);
10  console.log("Addition:", num1 + num2);
11  console.log("Subtraction:", num1 - num2);
12  console.log("Multiplication:", num1 * num2);
13  console.log("Division:", num1 / num2);
14  console.log("Modulus:", num1 % num2);
15  console.log("subhashree jena");
16
```

Output:-

```
PS C:\Users\IT-42\Desktop\VMERN stack> node practical1.js
Welcome to Node.js!
Number 1 = 20
Number 2 = 5
Addition: 25
Subtraction: 15
Multiplication: 100
Division: 4
Modulus: 0
subhashree jena
```

d.

Create and run a simple program using variables and functions.

Code:-

```
JS practical1b.js > ...
1   let num1 = 20;
2   let num2 = 5;
3
4   function add(a,b){
5     |   return a + b;
6   }
7   function mul(a,b){
8     |   return a * b;
9   }
10  console.log("first number:",num1);
11  console.log("second number:",num2);
12
13  console.log("addition = ",add(num1 , num2));
14  console.log("multiplication = ",mul(num1 , num2));
15  console.log("subhashree jena");
```

Output:-

```
PS C:\Users\IT-42\Desktop\MERN stack> node practical1b.js
first number: 20
second number: 5
addition = 25
multiplication = 100
subhashree jena
```

e. Demonstrate use of operators and control statement

Code:-

```
JS practical1E.js > ...
1  let a = 20 ;
2  let b = 40 ;
3  console.log("\nComparison Operators");
4  console.log ("a > b : ", a > b);
5  console.log ("a < b : ", a < b);
6  console.log ("a == b : ", a == b);
7  console.log ("a != b : ", a != b);
8
9  console.log("\nIf- Else Statement ");
10 if (a > b) {
11     console.log("a is greater than b ");
12 } else {
13     console.log("b is greater than or equal to a ");
14 }
15 // Switch Statement
16 console.log("\nSwitch Statement ");
17 let day = 2;
18
19 switch (day){
20     case 1 :
21         console.log("Monday");
22         break;
23     case 2 :
24         console.log("Tuesday");
25         break;
26     case 3 :
27         console.log("Wednesday");
28         break;
29     default :
30         console.log("Invalid day");
31 }
32
33 console.log("subhashree jena");
34
```

Output:-

```
PS C:\Users\IT-42\Desktop\MERN stack> node practical1E.js
```

Comparison Operators

```
a > b : false
a < b : true
a == b : false
a != b : true
```

If- Else Statement

b is greater than or equal to a

Switch Statement

Tuesday

subhashree jena

f. Write programs using loops and template literals.

Code:-

```
JS practical1f.js > ...
1   let number = 8 ;
2   console.log(`multiplication table of $ {number}`);
3   for (let i = 1; i <= 10 ; i++)
4   {
5       console.log(`${number} x $ {i} = $ {number * i}`);
6   }
7   }
8   console.log("subhashree jena");
9
```

Output:-

[illegible]